

PRODUCT MANUAL

Installation, Operation & Redress Manual - ACT-PKR-4500 'Patriot' Retrievable Packer

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Product Family	ACT-PKR-4500
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Product Revision History (from engineering records)

Rev	Date	Description of Change	By
A	2023-06-12	Initial release	R. Callahan
B	2024-02-20	Slip retainer screw upgraded to Gr 8; torque note added	A. Gutierrez
C	2024-10-15	Added optional Aflas element kit -03-AF per ECO-2024-067	R. Callahan

1. GENERAL

This manual covers installation, operation and redress of the ACT-PKR-4500 packer. The ACT-PKR-4500 is a hydraulic set packer for 4-1/2 in. casing with a maximum OD of 3.780 in., a minimum bore of 1.938 in., a differential pressure rating of 10,000 psi and a temperature rating of 300 degF (HNBR) / 350 degF (Aflas). End connections are 2-7/8 in. EUE 8RD.

Personnel installing this equipment must be trained in completion operations and must read this manual in full before use. Only ACT-supplied redress parts may be used; substitution voids the product warranty and API monogram qualification status.

NOTE: Keep this manual with the tool shipping file. The as-shipped Bill of Materials, material certifications and hydrostatic test report are traceable by the tool serial number engraved on the top sub.

2. SAFETY

WARNING: Pressurized equipment hazard. Never apply pressure above the rated working pressure. Bleed all trapped pressure to zero before breaking any connection.

WARNING: The setting mechanism contains a shear device (shear ring ACT-PKR-4500-07, 2,950 +/- 150 psi) calibrated at the factory. Field substitution of shear hardware with non-ACT parts can cause premature setting or failure to set.

- Wear impact gloves when handling slips; wicker edges are sharp.
- Use certified lifting equipment; see Section 3 for tool weights.
- Elastomer elements degrade in sunlight - store tools indoors, off concrete, below 120 degF.
- Follow customer site H2S contingency plan in sour service.

3. STORAGE AND HANDLING

Store horizontally on wooden cradles with thread protectors installed. For storage longer than 12 months, redress elastomers before running. Rotate stock first-in, first-out by the element cure date stenciled on the shipping band.

Item	Value
Shipping weight	see packing list (typ. 210-580 lb by size)
Lift points	Certified sling on body only - never on element or slips
Storage temperature	40 - 120 degF, indoors
Elastomer shelf life	HNBR 5 yr, Aflas 7 yr from cure date

4. PRE-RUN INSPECTION

- Verify part number and serial number against the delivery ticket and well program.
- Confirm element compound matches well fluid and temperature (HNBR std; Aflas for high CO₂/H₂S or >300 degF).
- Gauge maximum OD and verify against casing drift.
- Check all set screws present and witness-marked.
- Verify shear device part number (shear ring ACT-PKR-4500-07, 2,950 +/- 150 psi) against the running design.
- Function-check bore with drift bar: 1.938 in. minimum.

5. RUNNING AND SETTING PROCEDURE

The following procedure assumes the ACT-PKR-4500 is run on 2-7/8 in. EUE 8RD tubing with a wireline-settable plug or standing valve available to plug the string. Review the well schematic, confirm casing weight range, and verify drift below the setting depth before running.

Step	Action	Verify
1	Visually inspect packer; confirm shipping bands intact and element free of cuts or weather checking.	QC checklist Form F-77
2	Verify thread protectors removed and connections clean. Apply API 5A3 modified thread compound to pin ends only.	Compound coverage 100% of engaged threads
3	Make up bottom sub connection to tubing string. Torque per TS-100: see Section 6 torque table.	Torque chart within min/max
4	Run in hole at max 2 ft/sec in casing; slow to 0.5 ft/sec through liner tops, nipples and doglegs > 6 deg/100 ft.	Running speed log
5	Space out and land tubing. Set slackoff per completion design (typically 4,000-8,000 lbf for 4-1/2 in. casing).	Weight indicator
6	Plug tubing (pump-out plug, wireline plug in nipple, or standing valve).	Plug set confirmed
7	Apply tubing pressure in 500 psi steps to 2,950 psi (+/- 150 psi). Hold 5 min at each step.	Setting pressure achieved
8	Confirm set: slack off 10,000 lbf, then pull 10,000 lbf over string weight. No movement = packer set.	Weight indicator +/- 2,000 lbf
9	Pressure test annulus per well program (do not exceed casing rating or packer differential).	Test chart retained
10	Retrieve or shear out plug, resume completion program.	Program sign-off

6. MAKE-UP TORQUE

Make up all connections to the values below (from ACT specification TS-100). Values assume API 5A3 modified compound, friction factor 1.0.

Connection	Size	Min (ft-lb)	Optimum (ft-lb)	Max (ft-lb)
EUE 8RD box x pin	2-7/8 in.	1,650	1,850	2,050
Top sub to mandrel (integral shoulder)	-	2,800	3,000	3,200
Bottom sub to mandrel	-	2,800	3,000	3,200
Slip retainer screws 1/4-20 Gr 8	-	9 ft-lb	10 ft-lb	11 ft-lb
Drag block set screws 3/8-16 Gr 8	-	23 ft-lb	25 ft-lb	27 ft-lb

7. RELEASING / REMOVAL

The ACT-PKR-4500 releases with straight pull only. Pull 40,000 lbf over string weight to shear the release ring; the element relaxes and slips retract as the packer collapses to running OD. Pull out of hole at max 2 ft/sec, slowing at each restriction. If the packer has been set longer than 24 months, expect up to 60,000 lbf release force due to element adhesion; do not exceed 80% of tubing string yield.

8. REDRESS PROCEDURE

Redress the packer after every retrieval using redress kit ACT-PKR-4500-RK. Redress must be performed by an ACT-trained technician in a clean shop environment.

Step	Redress Action
1	Disassemble in tool order per assembly drawing. Tag and lay out parts on a clean bench.
2	Discard all elastomers, back-ups and the used shear device. Never reuse elastomers.
3	Clean metal parts in solvent bath; do not bead-blast sealing surfaces.
4	Inspect mandrel seal surfaces: max allowable defect 0.005 in. deep; polish out minor scratches with 400 grit.
5	MPI (wet fluorescent) the mandrel, cones and subs each redress; reject any linear indication > 1/16 in.
6	Gauge slip wickers: replace slips if any wicker crest is flattened more than 50%.
7	Install new seals from kit ACT-PKR-4500-RK. Grease o-rings with silicone-free assembly lube.
8	Install new shear device (shear ring ACT-PKR-4500-07, 2,950 +/- 150 psi) - verify rating stamped on part.
9	Reassemble; torque set screws per TS-100 Table 6; witness-mark all screws.
10	Complete redress record Form F-82 and return tool to serviceable stock.

NOTE: Redress interval: after every run, or 12 months in storage, whichever occurs first.

9. TROUBLESHOOTING

Symptom	Probable Cause	Remedy
Packer will not set at rated pressure	Tubing leak; plug not holding	Pressure test string; reset plug
Premature set while running	Excessive running speed through restrictions; debris	Slow to 0.5 ft/sec; circulate hole clean before re-run
Set pressure high by >300 psi	Wrong shear device installed	Verify shear part no. against BOM
Leak in annulus test	Element damage during running; casing scale at set depth	Retrieve, redress, reset 2 joints higher
Will not release at rated pull	Debris settled above packer	Circulate/wash over; apply max allowable overpull per string design

Document Approval

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